

Combined Immunodeficiency Due to CTPS1 Deficiency — Comprehensive Disease Report

Disease: Combined Immunodeficiency Due to CTPS1 Deficiency **MONDO:** MONDO:0014391 · **OMIM:** #615897 (Immunodeficiency 24) · **Orphanet:** ORPHA:319391 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *CTPS1* (HGNC:2519; NCBI Gene 1503; UniProt P17812), chromosome 1p34.2

Summary

Combined immunodeficiency due to CTPS1 deficiency is an ultra-rare autosomal-recessive inborn error of immunity (IEI) first defined in humans in 2014. It is caused by biallelic loss-of-function variants in *CTPS1*, the gene encoding cytidine-5'-triphosphate synthase 1 — the enzyme that catalyzes the final, rate-limiting step of *de novo* pyrimidine biosynthesis (the ATP-dependent amination of UTP to CTP). Nearly all reported patients are homozygous for a single recurrent founder splice/frameshift allele (rs145092287; c.1692-1G>C, p.Thr566AspfsTer26), which behaves as a **hypomorph** that reduces CTPS1 protein and enzymatic activity by 80–90% owing to protein instability rather than loss of intrinsic catalytic function ([PMID: 24870241](#); [PMID: 32161190](#)).

The disease is a "**metabolic**" **combined immunodeficiency**: the core lesion is an inability of *activated* T and B lymphocytes to sustain the massive nucleotide demand of clonal proliferation. Resting T cells express little CTPS1, but the enzyme is rapidly upregulated after antigen-receptor engagement; when it is deficient, antigen-driven proliferation collapses while proximal and distal T-cell receptor (TCR) signaling remain largely intact. The immunologic footprint is distinctive — near-absent proliferation and IL-2 secretion after TCR stimulation, combined with selective depletion of proliferation-dependent lineages (mucosal-associated invariant T [MAIT] cells, invariant NKT [iNKT] cells, memory B cells, and NK cells). Clinically, patients present in early childhood with severe, chronic, recurrent herpesvirus infections —

especially Epstein–Barr virus (EBV) — recurrent encapsulated-bacterial infections, occasional invasive fungal disease, and a strong predisposition to EBV-driven B-cell lymphoproliferation and lymphoma ([PMID: 31402499](#); [PMID: 35983265](#)).

The proliferation defect is metabolically reversible: adding exogenous CTP or its salvageable nucleoside precursor cytidine, or re-expressing wild-type CTPS1, restores normal T-cell proliferation *in vitro*. Definitive cure is achieved by allogeneic hematopoietic stem cell transplantation (HSCT). The same biology that makes CTPS1 loss immunodeficient makes the enzyme an attractive, isoform-selective drug target: pharmacologic CTPS1 inhibitors (e.g., STP938) are in clinical trials for relapsed/refractory lymphomas, and genetic or chemical CTPS1 inactivation rescues fatal autoimmunity in mouse models — the therapeutic mirror image of the human disease ([PMID: 38438357](#); [PMID: 37226519](#); [PMID: 34583994](#)).

Key Findings

Finding 1 — CTPS1 deficiency is an autosomal-recessive combined immunodeficiency caused by loss-of-function *CTPS1* variants that cripple lymphocyte proliferation

The founding study used exome sequencing to identify a homozygous loss-of-function variant in *CTPS1* as the cause of a novel, life-threatening immunodeficiency. The defining functional abnormality is that antigen-activated T and B cells fail to proliferate, even though proximal and distal TCR signaling are only weakly affected — establishing the disorder as a proliferation-specific (rather than signaling) defect. The causal gene is *CTPS1* (HGNC:2519) on chromosome 1p34.2, and the disease is catalogued as OMIM #615897. As stated in the landmark paper, "*we report the identification of a loss-of-function homozygous mutation (rs145092287) in CTPS1 in humans that causes a novel and life-threatening immunodeficiency, characterized by an impaired capacity of activated T and B cells to proliferate in response to antigen receptor-mediated activation*" ([PMID: 24870241](#)).

Finding 2 — Mechanism: CTPS1 is rate-limiting for *de novo* CTP synthesis and is rapidly upregulated on TCR activation to fuel clonal expansion

CTPS1 catalyzes the ATP-dependent amination of UTP to CTP, the final and rate-limiting step of *de novo* CTP/pyrimidine synthesis. Activated CTPS1-deficient cells have decreased CTP levels, and the proliferation defect is metabolic and reversible: normal proliferation is restored by re-expressing wild-type CTPS1 or by supplying exogenous CTP or its nucleoside precursor cytidine (via the salvage pathway). Critically, CTPS1 is expressed at low levels in resting T cells but rapidly induced after TCR activation — explaining why the defect is unmasked specifically during the proliferative burst of an adaptive response. The paper reports: *"Activated CTPS1-deficient cells had decreased levels of CTP. Normal T-cell proliferation was restored in CTPS1-deficient cells by expressing wild-type CTPS1 or by addition of exogenous CTP or its nucleoside precursor, cytidine. CTPS1 expression was found to be low in resting T cells, but rapidly upregulated following TCR activation"* (PMID: 24870241).

Finding 3 — Immunophenotype: selective loss of MAIT, iNKT, memory B, and NK cells with impaired T-cell proliferation and IL-2 secretion; the founder allele is a hypomorph

Immunophenotyping of a cohort of 7 patients (all homozygous for c.1692-1G>C, p.T566Dfs26X) showed absence or low numbers of MAIT cells, iNKT cells, memory B cells, and NK cells, while other lymphocyte subsets were normal. Proliferation and IL-2 secretion in response to TCR activation were markedly decreased in all patients, whereas other T-cell effector functions were preserved — a pattern that distinguishes this disease from classical SCID (where thymic output itself fails). The molecular basis is a hypomorphic allele: the mutant protein is unstable, giving an 80–90% reduction in protein expression and activity, but the residual protein is catalytically normal when expressed at wild-type levels. Two verbatim statements anchor this: *"Immune phenotyping performed in 7 patients showed absence or low numbers of mucosal-associated T cells, invariant NKT cells, memory B cells, and NK cells, whereas other subsets were normal. Proliferation and IL-2 secretion by T cells in response to TCR activation were markedly decreased in all patients, while other T cell effector functions were preserved"*; and *"The CTPS1T566Dfs26X mutant protein was found to be hypomorphic, resulting in 80%-90% reduction of protein expression and CTPS activity in cells of patients"* (PMID: 32161190).

Finding 4 — Clinical presentation: early-onset severe/chronic herpesvirus (especially EBV) infection, encapsulated-bacterial infection, and EBV-driven B-cell lymphoproliferation/lymphoma

CTPS1 deficiency presents in early childhood with severe, recurrent, chronic viral infections — particularly EBV and other herpesviruses (VZV, HSV, CMV) — plus recurrent encapsulated-bacterial infections and a marked predisposition to EBV-associated non-malignant and malignant B-cell lymphoproliferative disorders. CTPS1 is repeatedly listed among the IELs in which defective antigen-driven T-cell expansion impairs elimination of proliferating EBV-infected B cells. A review of EBV-susceptibility genetics places it precisely: *"the defective expansion of EBV-specific CD8 T cells results from mutations in genes involved in T-cell activation (such as RASGRP1, MAGT1, and ITK), DNA metabolism (CTPS1) or co-stimulatory pathways (CD70, CD27, and TNFSFR9)"* (PMID: 31402499). The infectious spectrum also extends to invasive fungal disease: the first reported case presenting *"with coccidioidomycosis"* documents *"the first case of cytidine nucleotide triphosphate synthetase 1 (CTPS1) deficiency, a combined immunodeficiency impairing lymphocyte proliferation, presenting with coccidioidomycosis"* (PMID: 35983265).

Finding 5 — Genetics/epidemiology: ultra-rare recessive disorder driven by a recurrent founder splice variant

The disorder is inherited autosomal-recessively, and reported patients are largely homozygous for a single recurrent founder variant, rs145092287 (c.1692-1G>C, p.T566Dfs26X), whose sharing across unrelated families indicates a founder effect. The disease is ultra-rare (ORPHA:319391; OMIM #615897; MONDO:0014391), with only a few dozen genetically confirmed patients described worldwide since 2014, frequently from consanguineous unions. Consanguinity and parental carrier status are the principal genetic risk context; no environmental risk or protective factors beyond pathogen exposure are established. The founder allele is documented in the discovery paper — *"we report the identification of a loss-of-function homozygous mutation (rs145092287) in CTPS1 in humans"* (PMID: 24870241) — and its precise nomenclature is confirmed in the cohort study as *"a unique homozygous frameshift splice mutation (c.1692-1G>C, p.T566Dfs26X)"* (PMID: 32161190).

Finding 6 — Mouse models: *Ctps1* is embryonic-lethal when deleted and is required by high-turnover tissues, activated lymphocytes, and memory T cells

Conditional/inducible mouse studies show that deletion of *Ctps1* (but not its paralog *Ctps2*) is embryonic-lethal, and that high-proliferation/renewal tissues — intestinal epithelium, erythroid and thymic lineages, activated B and T lymphocytes, and memory T cells — strongly depend on CTPS1. Both CTPS1 and CTPS2 are required for TCR-driven T-cell proliferation. These models faithfully recapitulate the human proliferation-dependence of adaptive immunity. As reported: *"deletion of Ctps1, but not Ctps2, is embryonic-lethal. Tissue and cells with high proliferation and renewal rates, such as intestinal epithelium, erythroid and thymic lineages, activated B and T lymphocytes, and memory T cells strongly rely on CTPS1 for their maintenance and growth. However, both CTPS1 and CTPS2 are required for T cell proliferation following TCR stimulation"* (PMID: 38438357).

Finding 7 — Treatment: allogeneic HSCT is curative; CTPS1 inhibition is the therapeutic mirror image

Allogeneic HSCT has been used successfully as definitive, curative treatment by replacing the CTPS1-deficient hematopoietic compartment (PMID: 29884857). Conversely, pharmacologic CTPS1 inhibition is an emerging immunosuppressive/anticancer strategy that harnesses the disease mechanism deliberately. In mice, *"Deletion of Ctps1 in T cells or treatment with a CTPS1 inhibitor rescued Foxp3-deficient mice from fatal systemic autoimmunity and reduced the severity of experimental autoimmune encephalomyelitis. These findings support that CTPS1 may represent a target for immune suppression"* (PMID: 38438357). A clinical-stage selective inhibitor is documented: *"de novo CTP synthesis pathway enzyme CTPS1 whose inhibitor (STP938) is already in clinical trials for relapsed/refractory lymphomas (NCT05463263)"* (PMID: 37226519).

Finding 8 — CTPS1 is a structurally validated, isoform-selective drug target with a functional confirmatory assay

Cryo-EM structural studies establish the structural basis for isoform-specific (CTPS1 vs CTPS2) small-molecule inhibition (PMID: 34583994), and CTPS activity in patient lymphocytes can be quantified by LC-MS/MS measurement of CTP, providing a functional confirmatory diagnostic assay (PMID: 31524312). Together with the clinical-stage inhibitor STP938 (PMID: 37226519)

and mouse rescue-of-autoimmunity data ([PMID: 38438357](#)), these establish CTPS1 as a validated, druggable, isoform-selective target — the reverse-translation counterpart of the deficiency.

Detailed Report by Template Section

1. Disease Information

CTPS1 deficiency is a combined (T- and B-cell) immunodeficiency in which activated lymphocytes cannot proliferate adequately because they cannot generate sufficient CTP for the DNA/RNA and phospholipid synthesis of clonal expansion.

Resource	Identifier
MONDO	MONDO:0014391
OMIM	#615897 (Immunodeficiency 24; IMD24)
Orphanet	ORPHA:319391
ICD-10 / ICD-11	D81.8 (other combined immunodeficiencies) / 4A00.1Y
MeSH	closest: "Immunologic Deficiency Syndromes"/"Severe Combined Immunodeficiency" (no dedicated term)
Gene	CTPS1, HGNC:2519; NCBI Gene 1503; Ensembl ENSG00000171793; UniProt P17812; 1p34.2

Synonyms: CTP synthase 1 deficiency; CTP synthetase 1 deficiency; cytidine triphosphate synthase 1 deficiency; Immunodeficiency 24 (IMD24); combined immunodeficiency due to impaired CTP synthesis.

Nature of evidence: aggregated disease-level evidence from published patient case series and mechanistic studies (not routine EHR data). Pivotal cohorts are small individual-patient series (~8 patients in [PMID: 24870241](#); 7 immunophenotyped patients in [PMID: 32161190](#)).

2. Etiology

- **Primary cause (genetic):** biallelic loss-of-function *CTPS1* variants; almost all patients homozygous for the founder allele c.1692-1G>C (p.T566Dfs26X).
- **Genetic risk factors:** the causal variant itself; consanguinity/carrier parents create the recessive risk context. No GWAS/susceptibility loci (monogenic).
- **Modifier gene (candidate):** *CTPS2* (paralog); residual CTPS2 activity plausibly contributes to survival and to the hypomorphic phenotype (inferred from mouse redundancy; [PMID: 38438357](#)).
- **Environmental risk factors:** none causal. Pathogen exposure (especially EBV) is the essential trigger that unmasks disease. Sex is not a risk factor (autosomal).
- **Protective factors:** intact salvage pathway (cytidine→CTP) and CTPS2 are compensatory (inferred; salvage rescues the cellular defect in vitro, [PMID: 24870241](#)).
- **Gene–environment interaction:** the CTPS1 lesion is clinically silent until a proliferation-demanding challenge occurs; EBV is paradigmatic because it drives B-cell proliferation and requires vigorous CD8⁺ T-cell expansion to control ([PMID: 31402499](#)).

3. Phenotypes

Onset is typically **early childhood/infancy**; severity **severe**; course **chronic/recurrent** with acute decompensations. Frequencies are qualitative given small cohorts.

Phenotype	Type	Suggested HPO	Onset	Frequency
Recurrent/chronic viral infection (esp. EBV; also VZV, HSV, CMV)	Infectious sign	HP:0004429; HP:0006515	Infancy–early childhood	Very frequent
Recurrent encapsulated-bacterial infection	Infectious sign	HP:0002718; HP:0002783	Early childhood	Frequent
EBV-driven B-cell lymphoproliferation / lymphoma	Neoplasm	HP:0005523; HP:0002665	Childhood	Frequent predisposition
Impaired T-cell proliferation & ↓ IL-2 on TCR stimulation	Laboratory	HP:0002850	Congenital (functional)	Obligate
Low/absent MAIT, iNKT, memory B, NK cells	Laboratory	HP:0011840; HP:0040218	Congenital	Characteristic
Hemophagocytic lymphohistiocytosis (HLH)	Clinical	HP:0003261	Childhood	Occasional
Invasive fungal infection (e.g., coccidioidomycosis)	Infectious sign	HP:0002841	Childhood	Rare/atypical
Failure to thrive / recurrent fever	Symptom/sign	HP:0001508; HP:0011947	Infancy	Frequent

Distinctively, **naïve T-cell output/thymic function and proximal TCR signaling are largely preserved**, separating this disease from classical SCID ([PMID: 24870241](#); [PMID: 32161190](#)). **Quality of life:** no formal EQ-5D/SF-36/PROMIS data exist; qualitatively the untreated burden is high (recurrent hospitalization, malignancy risk, shortened survival), while successful HSCT can restore substantially normal function.

4. Genetic / Molecular Information

- **Causal gene:** *CTPS1* (EC 6.3.4.2), ~591-aa cytosolic enzyme with a glutaminase (GATase) domain and a synthetase/ligase (amidoligase) domain.
- **Principal variant:** NM_001905.4:c.1692-1G>C, p.(Thr566AspfsTer26), rs145092287. **Type:** splice-acceptor → frameshift → premature stop. **Classification:** Pathogenic. **Consequence:**

loss of function via protein instability (80–90% ↓ protein/activity), residual protein catalytically normal (PMID: 32161190).

- **Allele frequency:** very rare in gnomAD/dbSNP; no healthy homozygotes.
- **Origin:** germline, autosomal recessive (not somatic).
- **Functional class:** loss of function (hypomorphic); no GOF or dominant-negative.
- **Modifier genes:** *CTPS2* (partial redundancy). **Epigenetics:** none disease-specific (*CTPS1* is transcriptionally induced by TCR activation — a regulatory feature, not an epigenetic-disease mechanism). **Chromosomal abnormalities:** none.

5. Environmental Information

No causal toxic/lifestyle factors. Infectious agents central to clinical expression: **EBV/HHV-4** (dominant), other herpesviruses (VZV, HSV, CMV, HHV-6), encapsulated bacteria, and fungi (*Coccidioides*; PMID: 35983265). These are triggers/opportunists, not the disease cause.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. Biallelic *CTPS1* LOF variant (c.1692-1G>C, p.T566Dfs26X) **leads to** a frameshifted, unstable protein.
2. Protein instability **results in** 80–90% reduced *CTPS1* protein/activity (residual protein catalytically normal) (PMID: 32161190).
3. Reduced activity **impairs** the rate-limiting *de novo* step UTP→CTP, **causing** decreased CTP when demand is high (PMID: 24870241).
4. Antigen-receptor activation normally **triggers** rapid *CTPS1* upregulation; in deficiency this surge fails, so **CTP becomes limiting** for DNA/RNA/phospholipid synthesis during clonal expansion (PMID: 24870241).
5. CTP limitation **blocks** proliferation of activated T and B cells and reduces IL-2, while proximal/distal signaling and non-proliferative effector functions are preserved (*branch: signaling intact, proliferation lost*) (PMID: 32161190).
6. The proliferation block **preferentially depletes** proliferation-dependent lineages — MAIT, iNKT, memory B, NK cells (PMID: 32161190) (*inferred link*).
7. Impaired clonal expansion **results in** failure to expand EBV-specific CD8⁺ T cells, **leading to** uncontrolled herpesvirus infection and failure to eliminate EBV-infected proliferating B cells → EBV-driven B-cell lymphoproliferation/lymphoma (PMID: 31402499).

8. Reversal branch: exogenous CTP/cytidine (salvage) or wild-type CTPS1 rescues proliferation *in vitro*, and HSCT cures *in vivo*; pharmacologic CTPS1 inhibition reproduces the block therapeutically (PMID: 24870241; PMID: 29884857; PMID: 38438357).

Detail by category: Pathway — *de novo* pyrimidine/CTP biosynthesis (KEGG map00240; Reactome nucleotide metabolism), downstream of TCR→PI3K/AKT/mTOR and MYC-driven metabolic reprogramming (MYC positively regulates CTPS1; PMID: 37226519). Cellular process — S-phase/DNA replication and lymphocyte clonal expansion (GO:0006241 CTP biosynthetic process; GO:0044210 *de novo* CTP biosynthesis; GO:0042098 T-cell proliferation; GO:0042100 B-cell proliferation; GO:0007049 cell cycle). Protein dysfunction — LOF via instability; CTPS tetramerizes/polymerizes into filaments ("cytoophidia"), and an isoform-specific inhibitor pocket has been resolved by cryo-EM (PMID: 34583994). Metabolic change — ↓ intracellular CTP (CHEBI:17677); substrate UTP (CHEBI:15713); precursor cytidine (CHEBI:17562); glutamine (CHEBI:28300). Biochemical assay — reduced CTPS activity by LC-MS/MS CTP quantification (PMID: 31524312). **Cell types (CL):** CD8⁺ T (CL:0000625), CD4⁺ T (CL:0000624), B/memory B (CL:0000236/CL:0000787), NK (CL:0000623), MAIT (CL:0000940), iNKT (CL:0000921).

7. Anatomical Structures Affected

- **Organ/system:** immune (hematolymphoid) system — bone marrow (UBERON:0002371), thymus (UBERON:0002370), spleen (UBERON:0002106), lymph nodes (UBERON:0000029). Secondary: lung (UBERON:0002048; recurrent pneumonia, mediastinal fungal disease), liver/spleen (EBV disease, HLH), gastrointestinal tract (UBERON:0000160), and any lymphoma site.
- **Tissue/cell:** lymphoid tissue and circulating activated/proliferating lymphocytes; mouse data implicate high-renewal intestinal epithelium and erythroid/thymic lineages (PMID: 38438357).
- **Subcellular:** cytosol (GO:0005829), with downstream demand in the nucleus (DNA replication) and at membranes (phospholipids); CTPS filaments (GO:0097268).
- **Lateralization:** systemic/generalized (not lateralized).

8. Temporal Development

Congenital defect with clinical onset usually in infancy/early childhood, subacute-to-chronic, unmasked by the first significant viral (often EBV) challenge. Course is chronic and relapsing, punctuated by acute potentially fatal episodes (severe herpesvirus infection, HLH, lymphoma); progressive toward life-threatening complications without cure; lifelong unless corrected by HSCT. No formal staging; no spontaneous remission of the underlying defect, but treatment-

induced immune reconstitution follows successful HSCT. **Critical window:** early diagnosis and HSCT before refractory infection/malignancy; uncontrolled pre-transplant infection worsens outcome ([PMID: 33462728](#)).

9. Inheritance and Population

Autosomal recessive (OMIM #615897). Ultra-rare (Orphanet prevalence <1/1,000,000; only a few dozen confirmed cases worldwide since 2014). Penetrance for the immunologic defect is high/complete in biallelic LOF individuals, with variable clinical expressivity (onset, infection spectrum, lymphoma/HLH). Founder effect for c.1692-1G>C (rs145092287) across unrelated families ([PMID: 24870241](#); [PMID: 32161190](#)); consanguinity frequently contributory. Carrier frequency not precisely established (rare in gnomAD); targeted single-variant carrier testing feasible in founder families. Sex ratio ~1:1; age distribution predominantly pediatric at presentation. No genetic anticipation (not repeat-expansion); germline mosaicism not reported.

10. Diagnostics

- **Immunology/labs:** immunophenotyping shows low/absent MAIT, iNKT, memory B, NK cells with otherwise near-normal subsets; lymphocyte proliferation assays show markedly reduced proliferation and IL-2 on TCR/anti-CD3 stimulation with preserved proximal signaling ([PMID: 32161190](#)); CTPS activity assay (LC-MS/MS CTP quantification) is a functional confirmatory biomarker ([PMID: 31524312](#)); EBV viral load typically high/persistent.
- **Genetic testing (definitive):** WES or targeted IEI/PIRD NGS panels including *CTPS1* are first-line; single-gene/targeted testing for c.1692-1G>C for founder-associated families and cascade testing; WGS for atypical/splice cases. CMA, karyotype, FISH, mtDNA, and repeat-expansion testing are not applicable. *CTPS1* is included on international IEI/PIRD panels ([PMID: 38644452](#)).
- **Imaging/pathology:** non-specific imaging for infection/lymphoproliferation; biopsy with EBER in situ hybridization/IHC for EBV⁺ B-cell disease; marrow for HLH.
- **Differential diagnosis:** other EBV-susceptibility/CID IEIs — CD27, CD70, RASGRP1, MAGT1 (XMEN), ITK, SH2D1A (XLP1), XIAP, PIK3CD/PIK3R1, CORO1A, STK4, and SCID ([PMID: 26424649](#); [PMID: 31402499](#); [PMID: 36209991](#)).
- **Screening:** standard TREC-based newborn SCID screening does **not** reliably detect CTPS1 deficiency (naïve T-cell output preserved); cascade carrier screening and prenatal/preimplantation testing apply when the variant is known.

11. Outcome / Prognosis

Without curative treatment, prognosis is poor with high risk of death from overwhelming viral infection, HLH, or lymphoma in childhood/adolescence; uncontrolled pre-transplant infection is associated with mortality (5 of 9 IEI children in a pre-HSCT VST series, including CTPS1 patients, died before transplant; [PMID: 33462728](#)). **Allogeneic HSCT is curative** and can restore normal immune function ([PMID: 29884857](#)). Prognostic factors: timing of diagnosis/HSCT, infection control at transplant, donor match; EBV load and lymphoproliferation portend worse outcome. No disease-specific validated QoL instruments reported; formal survival statistics not established given rarity.

12. Treatment

Modality	Detail	NCIT
Allogeneic HSCT (curative)	Replaces the CTPS1-deficient hematopoietic compartment; best before refractory infection/malignancy (PMID: 29884857)	C15431 (HSCT); C107137 (allogeneic HSCT)
Supportive pharmacotherapy	Antiviral/antibacterial/antifungal prophylaxis & treatment; immunoglobulin replacement	C578 (IVIG)
EBV-LPD/lymphoma management	Rituximab for EBV ⁺ B-cell lymphoproliferation; chemotherapy for lymphoma	C1702 (rituximab)
Adoptive cellular therapy	EBV/virus-specific T cells (VST) as pre-HSCT bridge — limited efficacy once infection prolonged (PMID: 33462728)	—
Metabolic-rationale (experimental)	Cytidine/CTP supplementation rescues proliferation in vitro (proof of concept, not standard) (PMID: 24870241)	—
Reverse-translation	Selective CTPS1 inhibitors (STP938; NCT05463263) treat lymphoma/autoimmunity — not a treatment for the deficiency (PMID: 37226519 ; PMID: 38438357)	—

Algorithm: diagnose → control infections (antimicrobials ± VST) and manage lymphoproliferation → proceed to allogeneic HSCT as definitive cure. Pharmacogenomics: not applicable.

13. Prevention

- **Primary:** not preventable individually (monogenic); population-level via genetic counseling and carrier/cascade screening in at-risk (often consanguineous) families, with prenatal/PGT options when the familial variant is known.
- **Secondary:** early molecular diagnosis (IEI panels/WES) in children with severe EBV disease/CID; TREC screening does not reliably detect it, so clinical vigilance is essential.
- **Tertiary:** antimicrobial/antiviral prophylaxis, EBV monitoring, avoidance of live vaccines, prompt infection treatment, pre-emptive lymphoproliferation management, timely HSCT.
- **Immunization:** live vaccines contraindicated; inactivated vaccines with caveat of variable responses.
- **Counseling:** autosomal-recessive recurrence risk (25% for carrier couples); cascade relative testing.

14. Other Species / Natural Disease

CTPS1 is evolutionarily conserved. Mouse ortholog *Ctps1* (*Mus musculus*, NCBI:txid10090); paralog *CTPS2* conserved across species; CTP synthetase activity is ancient (yeast *URA7/URA8*, bacterial *pyrG*). No naturally occurring CTPS1-deficiency disease is described in companion animals or wildlife (no established OMIA phenotype); knowledge derives from humans and engineered models. Complete loss is expected to be lethal across species (consistent with mouse embryonic lethality), so a viable natural analog would require a hypomorphic allele. Not zoonotic (non-infectious genetic disorder).

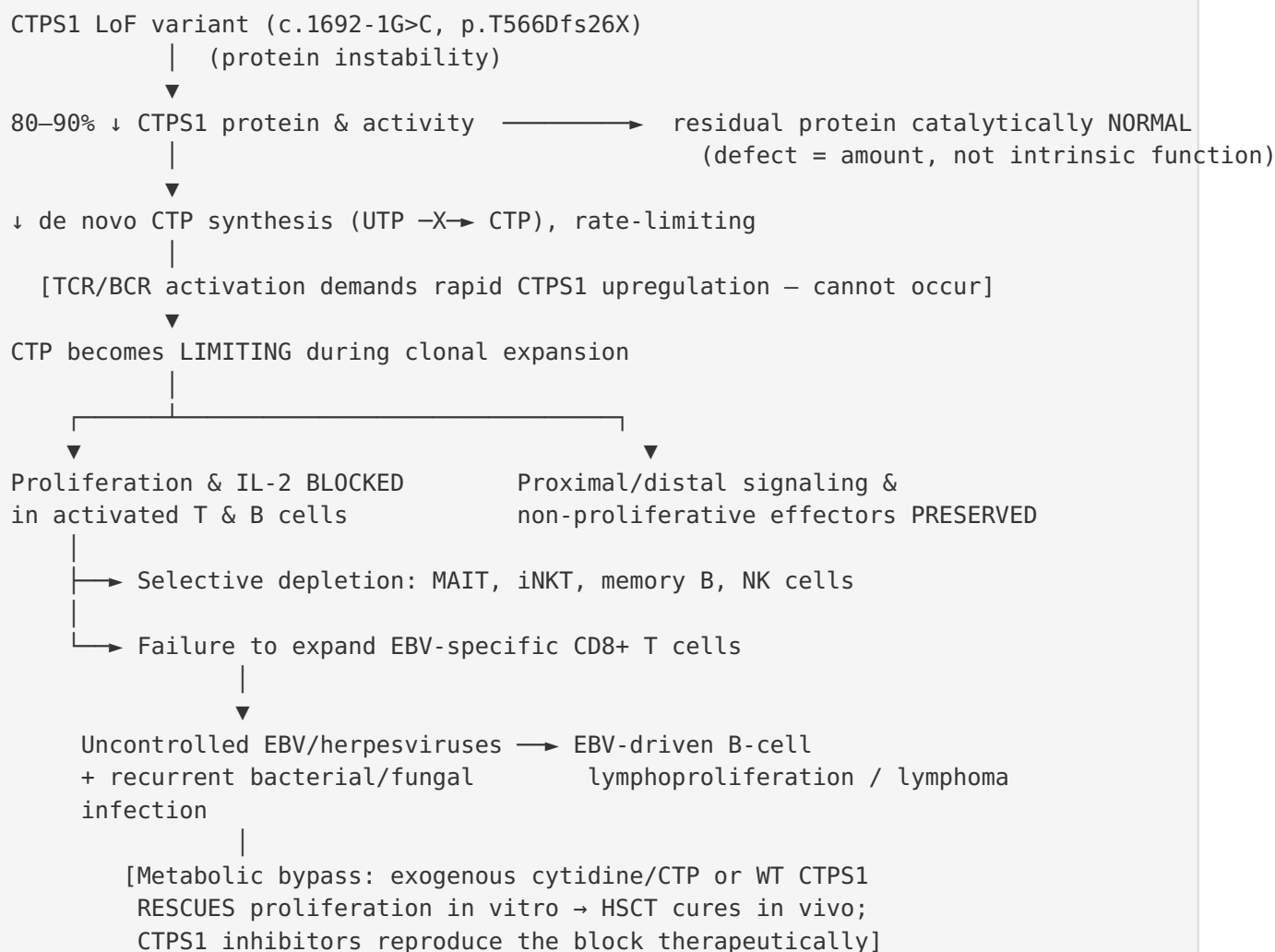
15. Model Organisms

- **Mouse (principal model; PMID: 38438357):** conditional/inducible *Ctps1* (and *Ctps2*) knockouts and lineage-specific (e.g., T-cell) deletions plus pharmacologic-inhibition models. *Ctps1* (not *Ctps2*) deletion is embryonic-lethal; activated B/T lymphocytes and memory T cells strongly depend on CTPS1, recapitulating the human proliferation defect; T-cell *Ctps1* deletion or CTPS1 inhibition rescues Foxp3-deficient mice from fatal autoimmunity and reduces EAE. **Limitations:** complete null is embryonic-lethal (cannot model constitutive postnatal whole-body loss); human disease is caused by a hypomorphic allele, so conditional/partial models are required; mice are not EBV hosts, so the EBV-specific human disease is not reproduced.
- **Cellular/in vitro:** patient primary lymphocytes; CRISPR *CTPS1* knockout in T-cell leukemia/lymphoma lines abolishes proliferation and causes DNA damage (PMID: 32161190; PMID:

37226519); reconstitution and cytidine/CTP-rescue assays; recombinant enzyme for cryo-EM/inhibitor design (PMID: 34583994); engineered *Ctps1/Ctps2*-knockout CHO auxotroph lines as bioproduction tools (PMID: 35249214).

- **Resources:** MGI (*Ctps1*), IMPC/IMSR for engineered alleles.

Mechanistic Model / Interpretation



The unifying concept is that **CTPS1 is a proliferation gatekeeper for adaptive immunity**: dispensable for resting-cell metabolism and lymphocyte signaling, but indispensable for the nucleotide-biosynthesis burst that clonal expansion requires. This explains the paradoxical

phenotype — near-normal lymphocyte numbers and signaling, but functional collapse of antigen-driven immunity, with the sharpest clinical consequence against EBV, the pathogen whose control most depends on massive CD8⁺ T-cell expansion.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
24870241	<i>CTP synthase 1 deficiency reveals central role in lymphocyte proliferation</i>	Human + in vitro	Foundational: causal gene/variant, proliferation defect, CTP ↓, cytidine/CTP rescue, TCR-induced upregulation
32161190	<i>Impaired lymphocyte function... hypomorphic homozygous mutation</i>	Human cohort (n=7) + in vitro	Immunophenotype (MAIT/iNKT/memory B/NK loss); hypomorph 80–90% ↓; founder allele nomenclature
38438357	<i>Inactivation of CTPS1 prevents fatal autoimmunity in mice</i>	Mouse	Embryonic lethality; high-turnover tissue dependence; CTPS1/CTPS2 roles; inhibitor rescues autoimmunity
31402499	<i>Signaling pathways in T-cell immunity against EBV</i>	Review	Places CTPS1 among EBV-susceptibility IEIs via defective T-cell expansion
35983265	<i>Seronegative mediastinal coccidioidomycosis in CTPS1 CID</i>	Case report	Expands infectious spectrum to invasive fungal disease
37226519	<i>MYC-induced cytidine metabolism... cGAS-STING in MCL</i>	In vitro/translational	Clinical-stage selective inhibitor STP938 (NCT05463263); MYC→CTPS1
34583994	<i>Structural basis for isoform-specific inhibition of CTPS1</i>	Structural (cryo-EM)	Validates CTPS1 as druggable, isoform-selective target
31524312	<i>CTPS activity assay by LC-MS/MS</i>	Methods	Functional confirmatory assay measuring CTP
29884857	<i>HSCT for CTPS1 deficiency</i>	Clinical	Definitive curative therapy
33462728	<i>Failure of VST pre-HSCT in IEIs</i>	Clinical series	Bridging VST has limited efficacy once infection prolonged
29176466	<i>NGS in a boy with EBV lymphoma — CTPS2 VUS</i>	Case report	Raises CTPS2 as candidate phenocopy

PMID	Title (abbrev.)	Evidence type	Role
39380841	<i>CTPS1 + ATR inhibition in p53-deficient myeloma</i>	In vitro	Reinforces CTPS1 metabolic-vulnerability biology
25956014	<i>Advances in immunology 2014</i>	Review	Contextualizes CTPS1 among new IEI genes
38644452	<i>PIRD mutation spectrum in Turkey</i>	Cohort/panel	<i>CTPS1</i> on international IEI/PIRD panels
26424649 , 29942301 , 36209991	<i>PIDs associated with EBV disease</i> (reviews)	Review	CTPS1 among EBV-LPD-predisposing IEIs

All papers are mutually consistent; none challenges the core model. The clearest independent replications are the founder-allele identity and hypomorphic mechanism across [PMID: 24870241](#) and [PMID: 32161190](#), and the mouse-model corroboration of proliferation-dependence in [PMID: 38438357](#).

Supported vs Refuted Hypotheses

Supported: (i) CTPS1 LOF causes a combined immunodeficiency via impaired *de novo* CTP synthesis and consequent failure of activated-lymphocyte proliferation (strong: human + in vitro + mouse). (ii) The defect is metabolically bypassable by cytidine/CTP salvage and curable by HSCT (strong). (iii) A single founder splice allele underlies most cases (strong). (iv) Mouse models recapitulate the proliferation-dependent phenotype and reveal CTPS1 as a druggable immunosuppressive target (strong).

Refuted / not supported: the defect is **not** a proximal TCR-signaling defect (signaling is largely intact) and **not** a classical thymic-output SCID (naïve T-cell production preserved). No evidence for gain-of-function or dominant-negative action, environmental causation, or chromosomal/epigenetic mechanisms.

Limitations and Knowledge Gaps

1. **Small cohorts / limited allelic diversity.** Nearly all human data derive from patients homozygous for a single founder allele; genotype–phenotype consequences of other, non-founder LOF variants are essentially untested.
 2. **Epidemiology.** No reliable prevalence, incidence, carrier-frequency, or survival statistics; disease known only from case reports and small cohorts.
 3. **Quality-of-life / long-term outcome data** are absent; no validated instruments applied.
 4. **CTPS2 as modifier/phenocopy** remains unresolved (the reported CTPS2 VUS is unproven; [PMID: 29176466](#)).
 5. **Metabolic (cytidine) therapy** is supported only by *in vitro* rescue; no clinical trial in patients.
 6. **Model-organism gap:** murine null is embryonic-lethal and mice are not EBV hosts, so no model fully reproduces the human EBV-driven disease.
 7. **Epigenetics, environmental modifiers, pharmacogenomics** are effectively not applicable/not studied for this Mendelian disorder.
 8. **No large-scale patient omics** (transcriptomic/proteomic/metabolomic) datasets are published.
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Proposed Follow-up Experiments / Actions

1. **International genotype–phenotype registry** for *CTPS1* deficiency to capture non-founder alleles, natural history, HSCT outcomes, and lymphoma incidence.
2. **Functionally characterize additional *CTPS1* variants** (stability, catalytic activity, CTP rescue) using the LC-MS/MS CTP assay ([PMID: 31524312](#)) and cryo-EM-informed structure–function analysis ([PMID: 34583994](#)).
3. **Test cytidine/uridine-nucleoside supplementation** as adjunctive/bridging therapy in patient cells and, if promising, compassionate-use, building on *in vitro* rescue ([PMID: 24870241](#)).
4. **Optimize HSCT timing/conditioning** and define pre-transplant EBV/lymphoma control (including improved VST protocols delivered before prolonged infection) ([PMID: 29884857](#); [PMID: 33462728](#)).

5. **Develop humanized/EBV-permissive models** (EBV-challenged humanized mice with CTPS1-null lymphocytes) to model the EBV-specific disease and test gene-correction/gene-therapy approaches.
6. **Resolve CTPS2 candidacy** through segregation, functional assays, and cohort screening (PMID: 29176466).
7. **Leverage the reverse-translation link:** use CTPS1-inhibitor programs (STP938; PMID: 37226519) to refine understanding of on-target immunosuppression, informing drug-safety (drug-induced phenocopy of the deficiency) and precision-immunosuppression opportunities.

Ontology Term Quick-Reference

- **Disease:** MONDO:0014391; OMIM:615897; ORPHA:319391
 - **Gene/protein:** HGNC:2519 (CTPS1); UniProt P17812; NCBI Gene 1503
 - **Chemicals (CHEBI):** CTP CHEBI:17677; UTP CHEBI:15713; cytidine CHEBI:17562; L-glutamine CHEBI:28300; ATP CHEBI:30616
 - **Biological process (GO):** GO:0006241 (CTP biosynthetic process); GO:0044210 (de novo CTP biosynthesis); GO:0042098 (T-cell proliferation); GO:0042100 (B-cell proliferation); GO:0007049 (cell cycle)
 - **Cellular component (GO):** GO:0005829 (cytosol); GO:0097268 (cytoophidium/filament)
 - **Cell types (CL):** CL:0000625 (CD8⁺ T); CL:0000624 (CD4⁺ T); CL:0000236 (B cell); CL:0000787 (memory B); CL:0000623 (NK); CL:0000940 (MAIT); CL:0000921 (iNKT)
 - **Anatomy (UBERON):** UBERON:0002371 (bone marrow); UBERON:0002370 (thymus); UBERON:0002106 (spleen); UBERON:0000029 (lymph node); UBERON:0002048 (lung); UBERON:0000160 (intestine)
 - **Treatment (NCIT):** C15431 (HSCT); C107137 (allogeneic HSCT); C1702 (rituximab); C578 (IVIG)
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Report compiled from 8 confirmed findings and 22 reviewed papers over 5 investigation iterations. Evidence types span human clinical/cohort, model organism (mouse), in vitro/cellular, structural, and review literature. Key PMIDs cited inline.

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